Appendix 3

Small Detector Approximation

$k_0 = 20;$
$X_0 = -5;$
$X_d = -3;$
$\delta = 0.5;$
$\delta_1 = 0.005;$

$n = 100;$
$Ti = -0.02;$
$Tf = 0.22;$
$dt = (Tf - Ti) / (n - 1);$

$tm[0] = Ti - dt;$
$tm[1] = Ti;$
$tm[n + 1] = Tf + dt;$

$Do[tm[i] = tm[i - 1] + dt, \{i, 2, n\}]$

$$\psi[0, x\_, t\_] := \left(\frac{\delta^2}{2\,\pi}\right)^{1/4} * \frac{Exp\left[-k_0^2 * \delta^2\right]}{\sqrt{\delta^2 + \frac{i*t}{2}}} * Exp\left[\frac{\left(2 * \delta^2 * k_0 + i * (x - X_0)\right)^2}{4 * \delta^2 + 2 * i * t}\right]$$

$$K[xf\_, tf\_, xi\_, ti\_] := \frac{1}{\sqrt{2 * \pi * i * (tf - ti)}} * Exp\left[\frac{i * (xf - xi)^2}{2 * (tf - ti)}\right]$$

$a[1] = Abs[\psi[0, X_d, tm[1]]] * Sqrt[\delta_1];$
$b[1] = \sqrt{1 - (a[1])^2};$

$Do[\alpha[0, j] = \psi[0, X_d, tm[j]], \{j, 1, n + 1\}]$

$Do[Do[\beta[i, j] = K[X_d, 0, X_d, tm[j] - tm[i]], \{j, i + 1, n + 1\}], \{i, 1, n\}]$

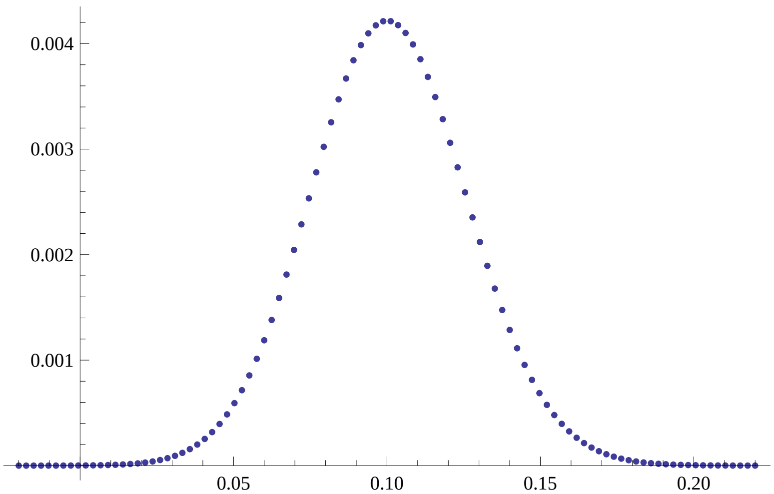
$$Do\Big[Do[\alpha[i, j] = (\alpha[i - 1, j] - \alpha[i - 1, i] * \beta[i, j] * \delta_1) / b[i], \{j, i + 1, n + 1\}];\ a[i + 1] = Abs[\alpha[i, i + 1]] * Sqrt[\delta_1];\ b[i + 1] = \sqrt{1 - (a[i + 1])^2}, \{i, 1, n\}\Big]$$

$$\text{Do}\left[\text{prob}[i\_] := \left(\prod_{k=1}^{i-1} b[k]^2\right) * (a[i])^2, \{i, 2, n, 1\}\right]$$

```
prob[1] = (a[1])^2;

ListPlot[Table[{tm[i], prob[i]}, {i, n}], PlotRange -> All]
```

$$\text{tbar} = \frac{\sum_{j=1}^{n} (\text{tm}[j] * \text{prob}[j])}{\sum_{j=1}^{n} \text{prob}[j]}$$



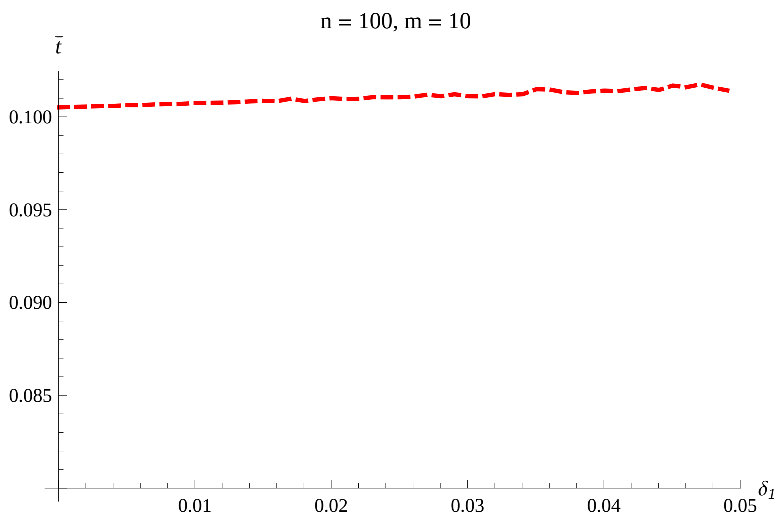

```
0.100794

.1 - tbar

-0.000793868
```

Random Position Approximations

```
m = 10; list2 = {}; X0 = -5;
Do[{{Do[acc[i] = 0; bcc[i] = 0, {i, 1, n}]; Do[{Do[tm[i] = tm[i - 1] + dt, {i, 2, n}],
    a[1] := Abs[ψ[0, RandomReal[{Xd - D1 / 2, Xd + D1 / 2}], tm[1]]] * Sqrt[D1];
    b[1] = Sqrt[1 - (a[1])^2]; acc[1] += a[1]; bcc[1] += b[1];
    Do[α[0, j] := ψ[0, RandomReal[{Xd - D1 / 2, Xd + D1 / 2}], tm[j]], {j, 1, n + 1}];
    Do[Do[β[i, j] := K[RandomReal[{Xd - D1 / 2, Xd + D1 / 2}], 0, RandomReal[
          {Xd - D1 / 2, Xd + D1 / 2}], tm[j] - tm[i]], {j, i + 1, n + 1}], {i, 1, n}];
    Do[Do[α[i, j] = (α[i - 1, j] - α[i - 1, i] * β[i, j] * D1) / b[i], {j, i + 1, n + 1}];
      a[i + 1] = Abs[α[i, i + 1]] * Sqrt[D1]; b[i + 1] = Sqrt[1 - (a[i + 1])^2];
      acc[i + 1] += a[i + 1]; bcc[i + 1] += b[i + 1], {i, 1, n}]}, {val, 1, m}];
   Do[{acc[i] = acc[i] / m; bcc[i] = bcc[i] / m}, {i, 1, n}]},
  Do[prob2[i] = (Product[bcc[k]^2, {k, 1, i-1}]) * (acc[i])^2, {i, 2, n, 1}];
  prob2[1] = (acc[1])^2; tbar1 = Sum[(tm[j] * prob2[j]), {j, 1, n}] / Sum[prob2[j], {j, 1, n}];
  list2 = Append[list2, {D1, tbar1}]}, {D1, .00005, .05, .001}];
a3 = ListLinePlot[list2, PlotRange -> All,
  PlotStyle → {Thick, Red, Dashed}, AxesOrigin → {0
    , 0.08}, PlotLabel → " n = 100, m = 10",
  AxesLabel → {Style["δ1", Italic, 10], Style["t̄", Italic, 10]}]
```



Non Directional & Dynamic Approach

```
list = {}; Do[list = Append[list, tm[i]], {i, 1, m}]; TList = Tuples[list, {n}];
```

```
Do[{Do[Tm[j] = TList[[i]][[j]], {j, 1, n}];
   Do[Do[If[Tm[i] ≠ Tm[j], {α[i, j] = (α[i - 1, j] - α[i - 1, i] * β[i, j] * δ1) / b[i]},
      {α[i, j] = 0}], {j, i + 1, n + 1}];
    a[i + 1] = Abs[α[i, i + 1]] * Sqrt[δ1]; b[i + 1] = Sqrt[1 - (a[i + 1])^2], {i, 1, n}];
   Do[prob[i, j] = (Product[b[k]^2, {k, 1, j - 1}]) * (a[j])^2, {j, 2, n, 1}];
   prob[i, 1] = (a[1])^2 }, {i, 1, Length[TList]}];
```

$$tbar = \frac{\sum_{i=1}^{Length[TList]} \sum_{j=1}^{n} (TList[[i]][[j]] * prob[i, j])}{\sum_{i=1}^{Length[TList]} \sum_{j=1}^{n} prob[i, j]}$$

```
0.100508
```

Directional & Dynamic Approach

```
Tlist = {}; Do[Tlist = Append[Tlist, Sort[TList[[i]]]], {i, Length[TList]}];
Tlist = DeleteDuplicates[Tlist];
```

```
Do[{Do[Tm[j] = Tlist[[i]][[j]], {j, 1, n}];
  Do[Do[If[Tm[i] ≠ Tm[j], {α[i, j] = (α[i - 1, j] - α[i - 1, i] * β[i, j] * δ1) / b[i]},
     {α[i, j] = 0}], {j, i + 1, n + 1}];
   a[i + 1] = Abs[α[i, i + 1]] * Sqrt[δ1]; b[i + 1] = Sqrt[1 - (a[i + 1])^2], {i, 1, n}];
  Do[prob[i, j] = (Product[b[k]^2, {k, 1, j - 1}]) * (a[j])^2, {j, 2, n, 1}];
  prob[i, 1] = (a[1])^2 }, {i, 1, Length[Tlist]}]
```

$$tbar = \frac{\sum_{i=1}^{Length[Tlist]} \sum_{j=1}^{n} (Tlist[[i]][[j]] * prob[i, j])}{\sum_{i=1}^{Length[Tlist]} \sum_{j=1}^{n} prob[i, j]}$$